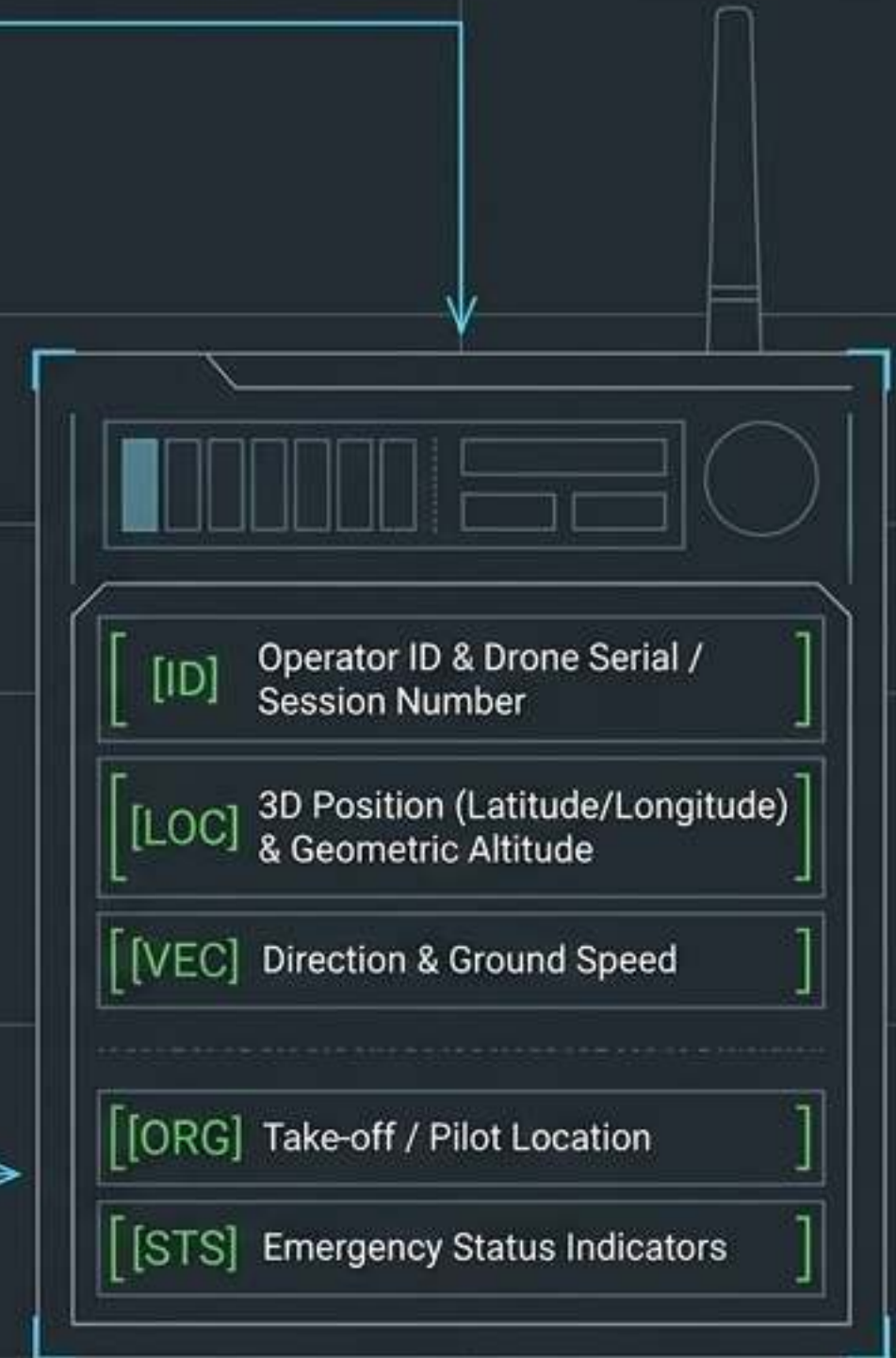
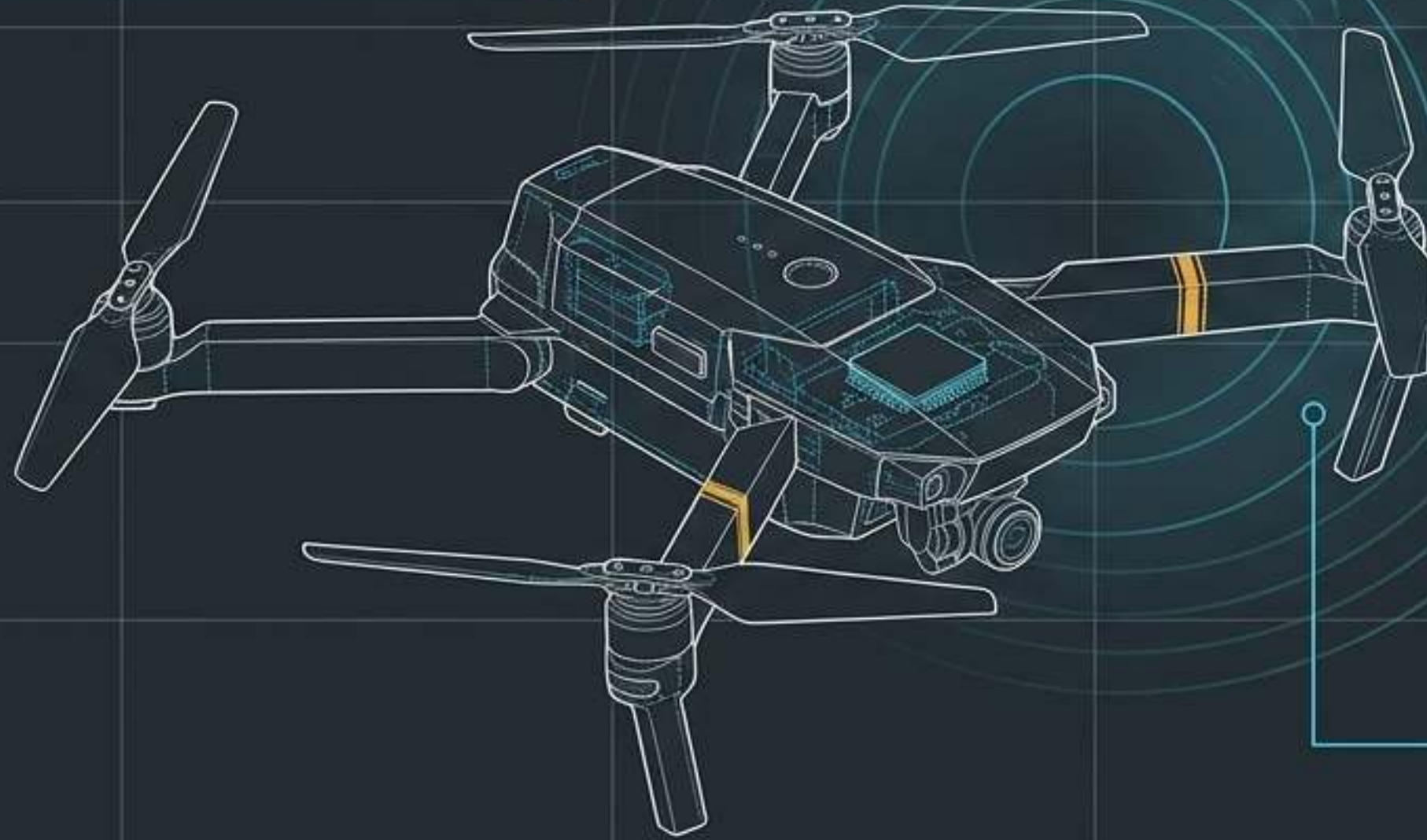


Remote ID Functions as a Digital License Plate for the Airspace

Direct Remote Identification (DRI) is the mandatory, continuous MAC-layer broadcast of UAS identification, location, and performance data to ground observers.



Trans-Atlantic Regulatory Frameworks Demand Continuous Compliance

FAA 14 CFR Part 89 (US)

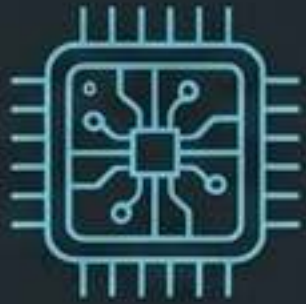
- **Scope:** Applies to all registered drones (>0.55 lbs).
- **Certification:** Requires FAA-accepted Declaration of Compliance (DoC) and an approved Means of Compliance (MoC).
- **Accountability:** Mandates continuous manufacturer audits and strict 15-day defect notification procedures.

Shared
Reliance:
MAC-Layer
Broadcasting

EASA & UK CAA (Europe)

- **Scope:** Mandatory Direct RID via WiFi/Bluetooth.
- **Data Structure:** Utilizes a standardized multi-part RID (Country Identifier Code + Public Code + Private Key).
- **Future Ambition:** Transitioning toward Hybrid RID, combining direct broadcast with continuous internet-connected Networked RID.

Operators Must Choose Between Three Explicit Compliance Pathways

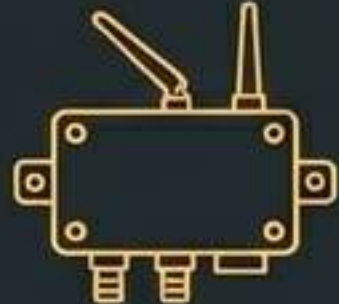


Standard Remote ID (Integrated)

Hardware: Built directly into the UAS by the manufacturer.

Broadcast Data: Drone ID + Drone Location + Control Station Location.

Operational Limit: Full operations permitted; critical for BVLOS (Beyond Visual Line of Sight) readiness.



Broadcast Module (Retrofit)

Hardware: Aftermarket retrofit attached externally to legacy non-RID drones.

Broadcast Data: Drone ID + Drone Location + Takeoff Location (Does not transmit control station).

Operational Limit: Strictly limited to Visual Line of Sight (VLOS) operations.



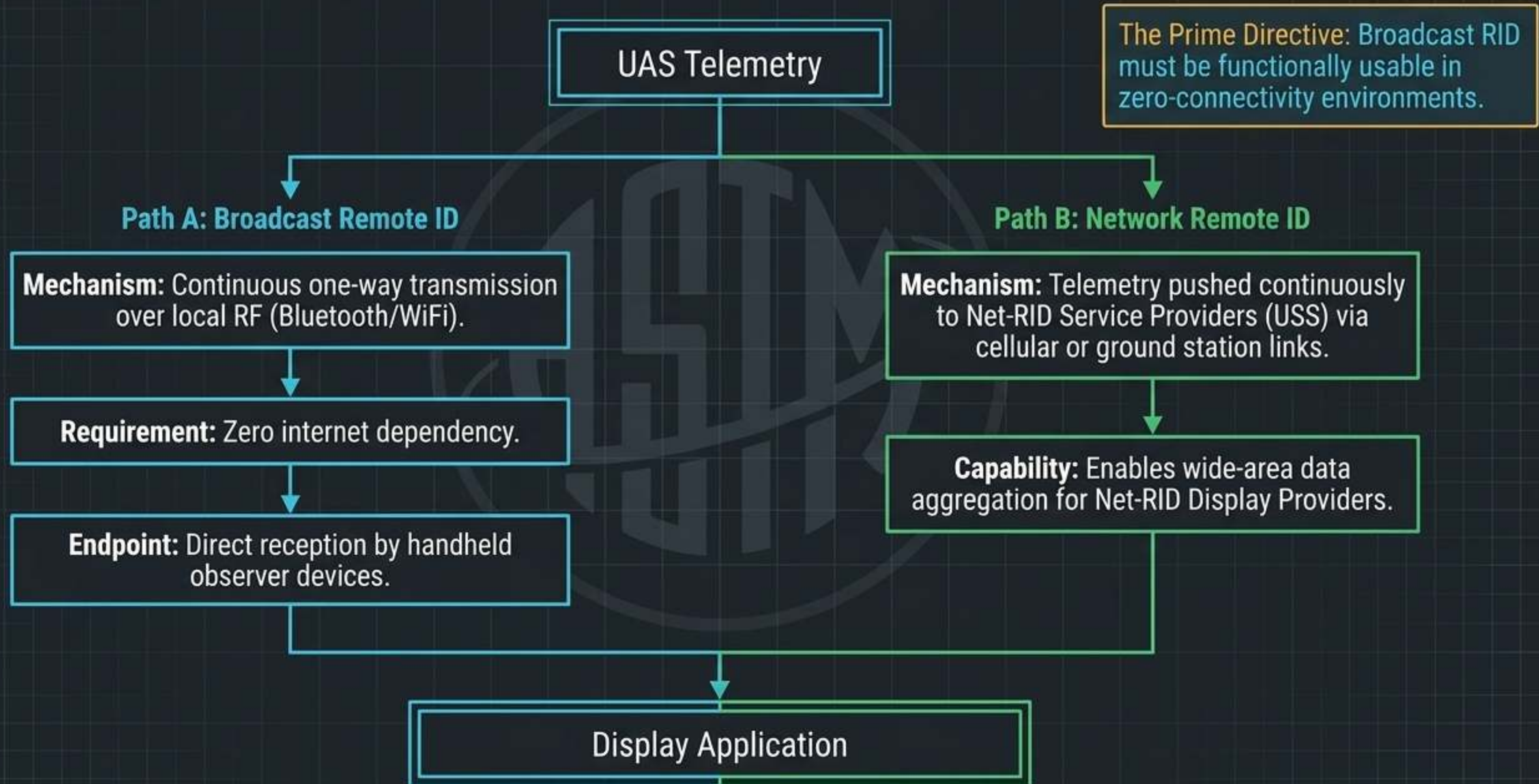
FRIA (FAA-Recognized Identification Area)

Hardware: No Remote ID equipment required.

Broadcast Data: None.

Operational Limit: VLOS only, geographically locked to approved Community-Based Organizations (CBOs) or Educational zones.

ASTM F3411-22a Dictates the Global Interoperability Architecture



Physical Layer Transmission Modalities Dictate Payload Capabilities



Bluetooth LE 4.x
(Legacy Advertising)

Payload: 25 bytes |
Update Rate: ~333ms

Mechanism: Rotational. Severe payload limits require rotating through message types (Basic ID → Location → System) rapidly to hit the 1.0Hz regulatory target. **Very High** consumer compatibility.



Bluetooth 5.x
(Extended)

Payload: 255 bytes

Mechanism: Unified Message Pack. Broadcasts all fields simultaneously. **Moderate but growing compatibility.**



WiFi NaN
(802.11v/z)

Payload: Variable

Mechanism: Discovery Protocol. **High compatibility on Android, severely limited on iOS.**



WiFi Beacon
(802.11 Mgmt)

Payload: Variable

Mechanism: Broadcast SSID Vendor Element. Embeds RID data directly into wireless management frames.

Very High compatibility.

OpenDroneID Compiles the Standard into Deployable C-Libraries

Telemetry Integration

Raw data encapsulated via dedicated MAVLink messages from professional flight controllers (e.g., Pixhawk / CubePilot).

Protocol Translation

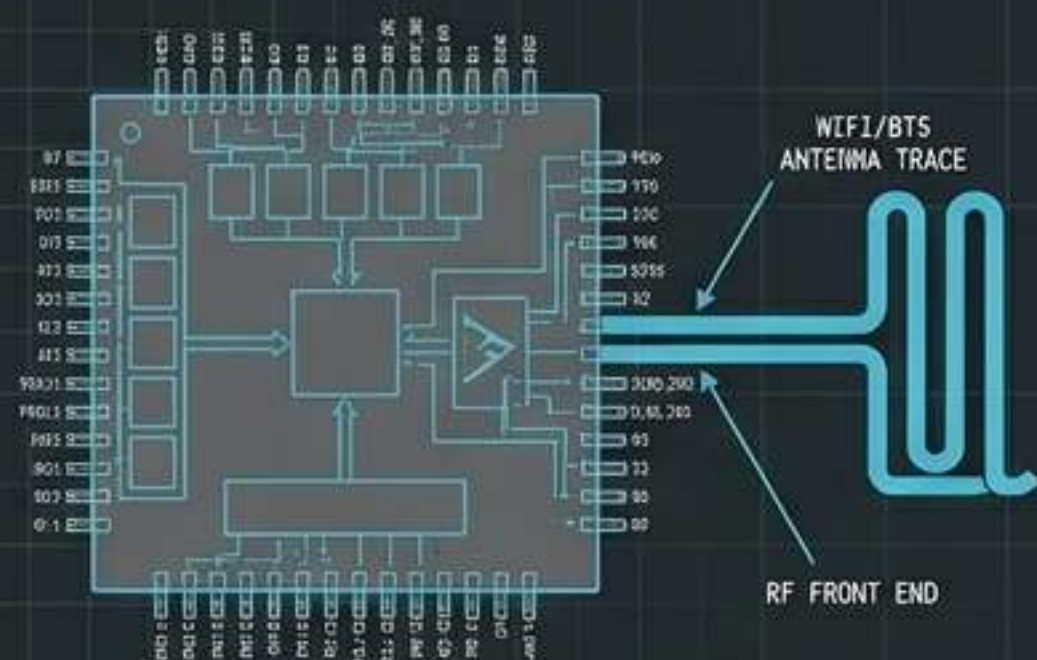
OpenDroneID Core C-library translates ASTM/ASD-STAN regulatory specs into nominal C-structures.

Payload Encoding

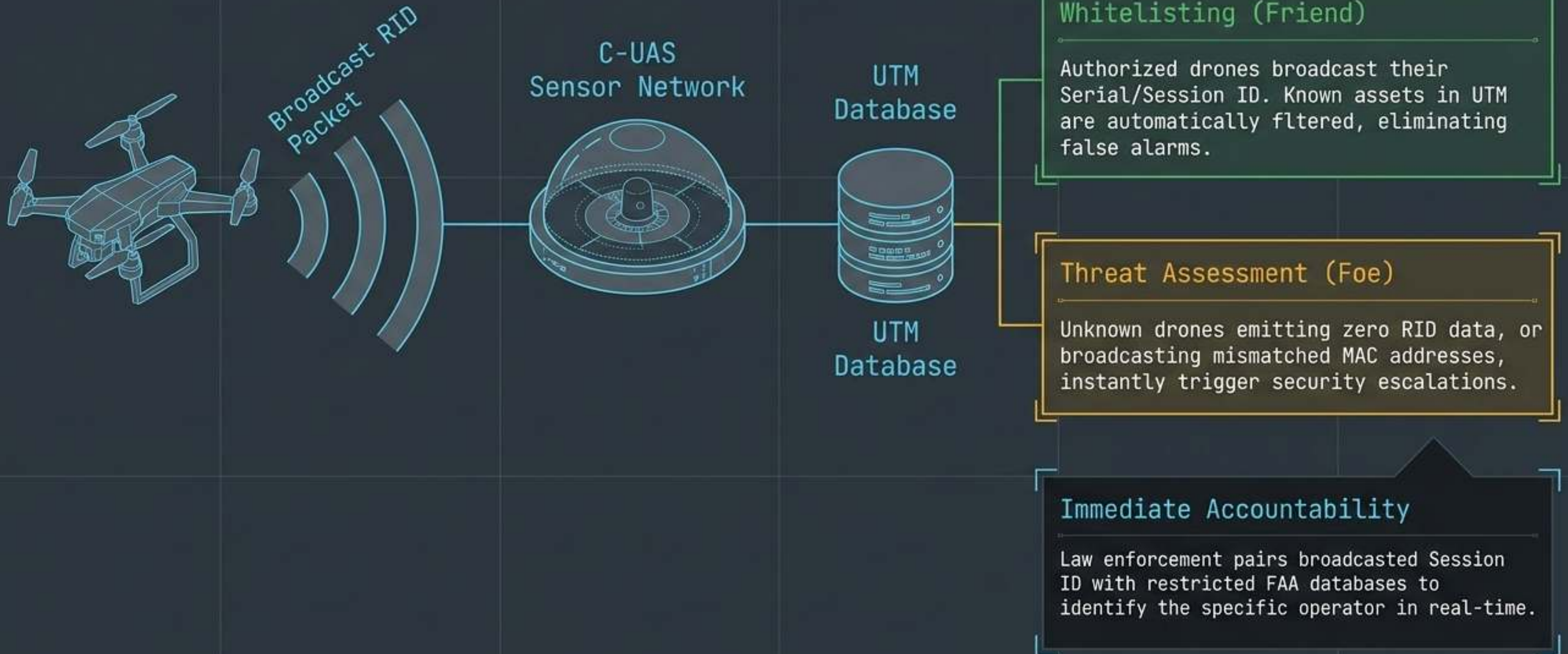
Packages translated structures into **[Message Packs]** (Basic ID, Location, System).

Hardware Reference Execution

Code optimized and executed on microcontrollers like the **ESP32-S3/C3** (providing native WiFi/BT5) or **nRF52840** chipsets for MAC-layer broadcast.



C-UAS and UTM Networks Consume RID to Separate Friend from Foe



The Baseline Standard Leaves a Critical Void in Cybersecurity



The Privacy Shield

A license plate, not a driver's license.

The broadcast contains **Serial or Session IDs**, not direct Personally Identifiable Information (PII).

Accessing PII requires secure law enforcement **database queries**.



The Spoofing Threat Vector

ASTM F3411-22a leaves **broadcast authentication** unspecified or demands an **active internet connection** to verify signatures. This creates massive gaps for **offline spoofing**.

Exploitation Tactics:

- >_ **Spoofing**: Adversaries inject fake compliance packets via off-the-shelf SDRs.
- >_ **Replay Attacks**: Malicious actors capture a legitimate flight's RID packet and re-broadcast it later in restricted airspace to trigger false enforcement actions.

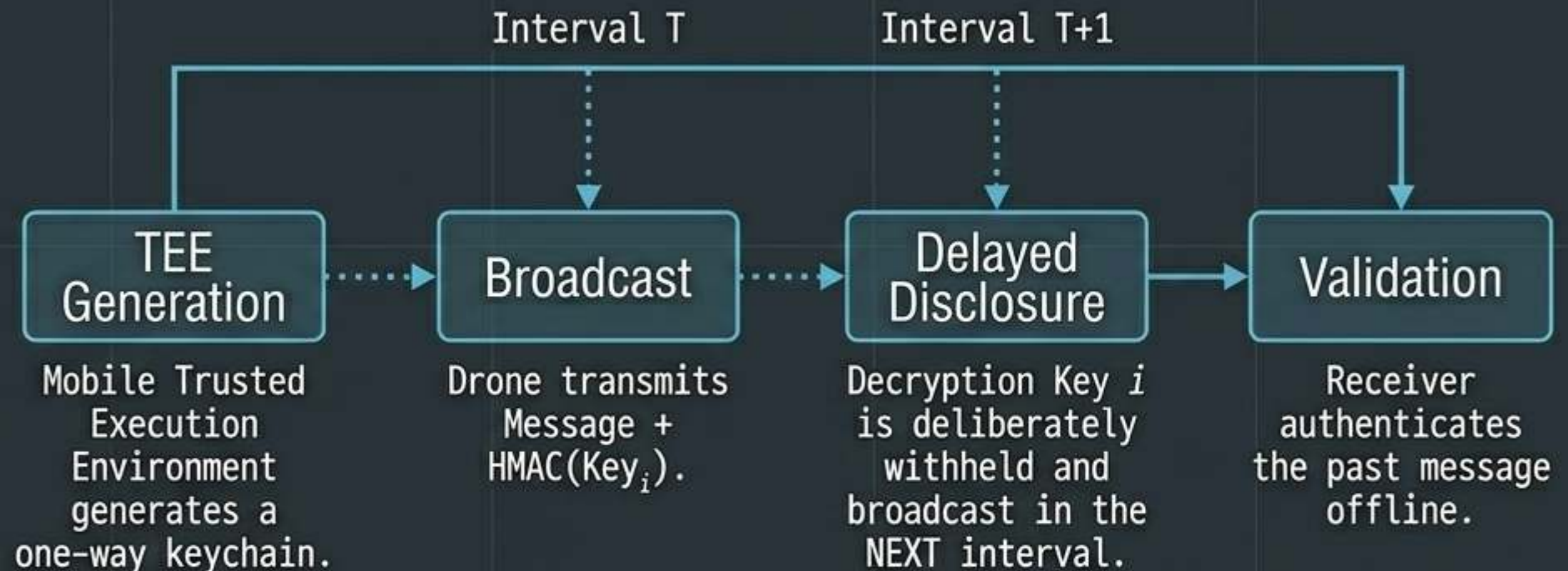
TBRD Mitigates Spoofing Through Delayed Key Disclosure

The Computational Barrier:

Traditional ECC digital signatures are too heavy for 1Hz broadcasts (139 bytes payload, 1.2s compute time).

The Protocol Fix:

TBRD uses symmetric cryptography (SHA-256 HMAC) and delayed key disclosure to achieve offline authentication.



The Result:

Only 68 bytes of overhead and 10ms of compute time.

Total Airspace Transparency is the Prerequisite for Autonomy

[GLOBAL_DEADLINES]

Mapping the alignment of FAA operational enforcement deadlines with impending EASA 2026 mandates to force ecosystem-wide adoption.

[AUDITS_&_COMPLIANCE]

Part 89 requires strict record retention for MoC submissions. Failure to maintain active product support or issue 15-day defect notices results in **immediate FAA rescission** of the DoC.

[THE_AUTONOMOUS_HORIZON]

Scaled, decentralized drone swarms and high-density UTM integration are impossible without the trust provided by **authenticated, ubiquitous Remote ID**.

Compliance is the fundamental infrastructure for the future of flight.

FAA Operational
Enforcement

EASA 2026
Mandates

Continuous
Manufacturer Audits

Total UTM
Integration